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AI CAREERS FOR WOMEN

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EDUPULSE: AI-DRIVEN REAL-TIME STUDENT FEEDBACK ANALYTICS FOR SMART LEARNING

EDUCATIONAL ANALYTICS & NLP

Project Domain

Excel + Power BI + Copilot /AI Chat bot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. PROBLEM STATEMENT

In the educational domain, student feedback plays an important role in evaluating the effectiveness of teaching and learning processes. Colleges and universities regularly collect feedback from students regarding various aspects which helps the institutions to understand whether the teaching methods used by faculty members are effective and whether students are able to achieve the expected learning outcomes. However, most of the feedback collected is unstructured in nature, especially in the form of written comments and multiple rating variables, which makes the analysis process difficult and time-consuming.

Significance of the Problem

Analyzing student feedback is significant for improving teaching quality and academic performance. When feedback is properly analyzed, it helps faculty members identify their strengths and weaknesses and make necessary improvements in their teaching methods. It also helps academic coordinators and management to monitor faculty performance, maintain education standards, and ensure better learning outcomes for students. Proper feedback analysis supports data-driven decision making, which is essential in modern education systems where quality assurance and continuous improvement are important.

Gap Identified

The educational institutions collect large amounts of feedback data but there is a lack of integration between feedback collection and data analysis tools. In many cases, feedback is stored but not analyzed using modern tools like Excel analytics, Power BI dashboards, or AI-based sentiment analysis. Because of this gap, useful insights such as low teaching clarity, weak interaction, or negative student opinion are not identified properly. There is a need for a structured system that can organize feedback data and generate meaningful insights automatically. Modern tools such as Excel, Power BI, and AI-based text analysis can help convert raw feedback data into meaningful insights, charts, and recommendations.

Stakeholders Benefited

Faculty members, Academic coordinators, College management, Students, and Educational institutions benefit from proper feedback analysis because it helps in improving teaching effectiveness and maintaining academic quality.

Value Delivered

By using data analytics, Excel, Power BI, and AI tools, the student feedback data can be converted into structured reports, visual dashboards, and automated summaries. This helps in identifying strengths, weaknesses, and areas for improvement in teaching. The system provides clear insights, better decision-making support, and continuous improvement in teaching and learning process.

4. OBJECTIVES

- To collect, clean and preprocess the feedback data containing multiple feedback variables by removing missing values, checking data consistency, and standardizing rating scales using Microsoft Excel.
- To analyze the feedback dataset using Excel formulas, pivot tables, and statistical methods to identify patterns in teaching performance and student satisfaction.
- To apply basic sentiment analysis on feedback comments to classify responses using AI tools of chat bot assistance.
- To build an interactive Power BI dashboard to visualize faculty performance, course-wise feedback, rating trends, and improvement areas.
- To use AI tools such as Copilot to generate automatic summaries and teaching improvement suggestions based on feedback data.
- To document the complete project process, upload all files to GitHub, generate an analytical academic report and record a demonstration video for submission

5. METHODOLOGY / IMPLEMENTATION PLAN

SN	PHASE	DESCRIPTION
1.	Data Collection & Exploration	Create and obtain a student feedback dataset in Excel/CSV format containing multiple feedback variables. Understand field names, data types, and row count and convert them into structured text format

2.	Data Cleaning & Preprocessing	Use Microsoft Excel to clean the dataset by removing duplicate records, handling missing values, correcting data format and standardizing rating scales. Organize the data into proper columns such as Course Name, Faculty Name, Ratings, Feedback Text, Sentiment, and Recommendation to make the dataset ready for analysis.
3.	Analysis & AI Integration	Apply Excel formulas, Pivot Tables, COUNTIF, AVERAGE, and IF functions to analyze feedback ratings and identify teaching performance trends. Use AI tools such as Microsoft Copilot / AI Chat bot to perform sentiment analysis on feedback text, generate summaries, and identify areas for teaching improvement. Compare different variables such as clarity, interaction, and subject knowledge to find key factors affecting overall rating.
4.	Visualization / Output	Create charts, pivot charts, and dashboards using Excel or Power BI to show faculty-wise ratings, course-wise feedback, sentiment distribution, and improvement areas. Use KPI cards, slicers, and bar charts to present the results in visual format. Document all AI prompts used and store the generated summaries along with the dashboard outputs.
5.	Documentation & Submission	Prepare the complete project report including problem statement, objectives, dataset description, analysis, dashboard results, and conclusion. Upload dataset, Excel file, dashboard, and report in GitHub repository. Record a demonstration video explaining dataset creation, analysis, AI usage, and dashboard visualization as per submission guidelines.

6. RESOURCE AND TIMELINE

A. Hardware & Software Requirements

- Laptop or PC (Minimum 4 GB RAM)
- Operating System: Windows 10

- Microsoft Excel (Microsoft 365)
- Power BI Desktop
- Microsoft Copilot
- AI Chat bot –Chat GPT
- Internet Connection- required for Copilot and AI Chat bot access
- Web browser — Google Chrome

B. Dataset

- **Source:** Student feedback dataset created for project purpose and simulated using Excel. The dataset represents feedback collected from students regarding teaching performance, course quality, and learning experience in higher educational institutions.
- **Format:** Excel / CSV dataset created with structured columns containing ratings, feedback text, sentiment, and improvement areas.
- **Dataset Type:** Structured student feedback dataset – Course Name, Faculty Name, Teaching Clarity, Content Quality, Interaction, Subject Knowledge, Communication Skill, Use of Technology, Motivation Level, Overall Rating, Feedback Text, Sentiment, Improvement Area, Recommendation, and Submission Date.
- **Size of Dataset:** 500 rows and multiple feedback variables to enable proper analysis, dashboard creation, and AI-based summarization.
- **Nature of Data:** Mixed data type including numerical ratings, categorical variables, and text feedback comments used for sentiment analysis.
- **Purpose of Dataset:**
  The dataset is used to analyze teaching effectiveness, identify strengths and weaknesses in teaching, perform sentiment analysis on student feedback, and generate visual dashboards using Excel, Power BI, and AI tools.
- **Usage in Project:**
  The dataset will be used for data cleaning, preprocessing, statistical analysis, AI-based feedback summarization, and dashboard visualization to develop a Student Feedback & Teaching Improvement Analyzer

C. Project Timeline

PHASE	TASK	TIME COMPLETION
Phase 1	Data collection ,exploration and structure design	Week 1
Phase 2	Data cleaning and preprocessing and validation	Week 2
Phase 3	AI integration and analysis-Excel formula, Copilot	Week 2–3
Phase 4	Dashboard creation. insight and review	Week 3
Phase 5	Report writing, GitHub upload, and video recording and submission	Week 4

7. EXPECTED OUTCOMES/CONCLUSION

This project is expected to produce a structured student feedback analysis system using Excel, Power BI, and AI tools to evaluate teaching performance and improve the quality of education. A clean and well-organized dataset containing multiple feedback variables such as teaching clarity, content quality, interaction, subject knowledge, communication skill, motivation level, and overall rating will be prepared after data cleaning and preprocessing.

The project will generate meaningful insights from the feedback dataset using Excel formulas, pivot tables, and statistical analysis. Sentiment analysis will be performed on feedback comments to classify responses as positive, neutral, or negative. These results will help in identifying strengths and weaknesses in teaching methods and understanding student satisfaction levels.

An interactive dashboard will be created using Excel or Power BI to visually present faculty-wise performance, course-wise ratings, sentiment distribution, and improvement areas. The dashboard will help academic coordinators and faculty members to easily understand feedback results and take necessary actions for teaching improvement.

AI tools such as Copilot or AI Chatbot will be used to generate automatic summaries of feedback comments and suggest possible improvements in teaching methods. This will demonstrate how Artificial Intelligence can support educational analytics and decision making.

The final outcome of the project will include a cleaned dataset, analysis report, dashboard visualization, AI-generated summaries, GitHub repository, and project documentation. The developed feedback analyzer can be reused by educational institutions to monitor teaching effectiveness, improve academic quality, and support continuous improvement in the teaching-learning process.

8. REFERENCES

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